New Challanges In The Kinetic Theory of Complex Fluids

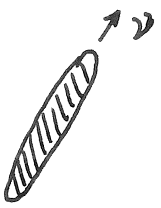
In this talk I'll focus my attention on the particular sub class of fluids known as COMPLEX FLUID. A complex fluid is a fluid where every molecular constituent is endowed with some sort of microstructural properties.

EXAMPLE (Gas Bubbles) Eeach molecular constituents will be a gas bubble, with a particular volume $v \in \mathbb{R}_{\geq 0}$. The additional microstructure in this case is the bubble's volume.



EXAMPLE (Calamitic Molecules) Each molecular constituents will here be a ROD-LIKE molecule with vanishing girth. Here the additional microstructure is the direction of the molecular axis, $v \in \mathbb{S}^2$. This is assuming that the calamitic molecule has no head to tail symmetry, if we have one head to tail symmetry then we can take $v \in \mathbb{RP}^2$.

$\frac{4}{3}\pi R = v$





EXAMPLE (Magnetic nano-particles) In a ferro fluid we regard the molecular constituents as magnetic nano-particle. Here the micro structure associated with each constituents is the vector $v \in \mathbb{R}^n$ representing the magnetic moment of each magnetic nano-particle.

If we hope to build a unified kinetic theory for complex fluid we first need to find a unified way to describe the molecular microstructure that encodes the complexity of the fluid. Our tool to do so will be CAPRIZ'S MICRO STRUCTOR (OR ORDER PARAMETER) MANIFOLD, introduced by CAPRIZ to describe from a phenomenological point of view the order

parameter associated with a complex fluid.

DEFINITION Given a manifold of dimension $\iota \in \mathbb{N}$, here denoted $\mathcal{M}$ and to be intended with a single parametrisation chart which is fixed a priori. Given also a Lie group action $A: \mathcal{M} \times SO(d) \to \mathcal{M}$ we call the tuple $(\mathcal{M}, A)$ a MICRO STRUCTURE (OR ORDER PARAMETER) MANIFOLD.

We also call a map $\nu: \mathbb{R}^d \to \mathcal{M}$ an order parameter manifold if,

1. $\nu(\underline{x} + \underline{b}) = \underline{\nu}(\underline{x}) \quad \forall \underline{b} \in \mathbb{R}^d$
2. $\nu(Q\underline{x}) = A(\nu(x), Q) \quad \forall Q \in SO(d)$.

The nice property of the MICRO STRUCTURE MANIFOLD is that by its property we can characterise certain behaviour of complex molecule interactions.

Let us consider one molecular constituent with generalised Lagrangian coordinates $(x, \nu) \in \mathbb{R}^d \times \mathcal{M}$ and corresponding generalised Lagrangian velocities $(\underline{v}, \omega) := (\dot{x}, \dot{\nu}) \in \mathbb{R}^d \times \underbrace{\mathbb{R}^\iota}_{T_\nu \mathcal{M}}$. We will assume that the dynamics of this single molecular constituent is given by the Lagrangian $\mathcal{L}(x, \nu, \underline{v}, \underline{\omega}) = \frac{1}{2} m |v|^2 + \frac{1}{2} \omega^T B(\nu) \omega$ where $B(\nu)$ is a symmetric positive definite element of $\mathbb{R}^{\iota \times \iota}$ for every ν, trasforming as a COVARIANT RANK 2-tensor.

PROPOSITION If the group action associated with the MICRO STRUCTURE MANIFOLD is TRANSITIVE, i.e. $\forall \nu \in \mathcal{M}$ and $\nu' \in \mathcal{M}$ it $\exists Q \in SO(d)$ such that $\nu = A(\nu', Q)$, then $B(\nu) = B$.

PROOF Fix ω and pick $\nu, \nu' \in \mathcal{M}$ $\frac{1}{2} m |v|^2 + \frac{1}{2} \omega^T B(\nu) \omega = \frac{1}{2} m |v|^2 + \frac{1}{2} \omega^T B(\nu') \omega \Rightarrow \omega^T B(\nu) \omega = \omega^T B(\nu') \omega \Rightarrow B(\nu') = B(\nu)$.

We will also consider binary interactions between two molecular constituents with coordinates in the phase space $\underbrace{(x_i, v_i, v_i, \omega_i)}_{\Gamma_i}$. In particular, we will assume that the dynamics of such interaction is governed by the lagrangian,

$$\mathcal{L} = \sum_{i=1}^{2} \frac{1}{2} m |v_i|^2 + \frac{1}{2} \underline{\omega}_i^T B(v_i) \underline{\omega}_i - W(x_1 - x_2, v_1, v_2).$$

NOETHER'S THEOREM Given an infinitesimal generator G for the symmetry group S of $\mathcal{L}$, we know that $\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\pi}} \cdot G\right) = 0$, with $\pi = (x_1, x_2, v_1, v_2)$.

COROLLARY Since the lagrangian $\mathcal{L}$ admits as symmetry S which has infinitesimal generator $G = (1,1,0,0)$ we then have that $\frac{d}{dt}(p_1 + p_2) = 0$. This is the conservation of linear momentum.

COROLLARY In $d=3$, the infinitesimal generator of the canonical action of $SO(d)$ is $q \mapsto q \times x_1$ and thus if we denote by $A_{v_i} q$ the infinitesimal generator corresponding to v_i evaluated at q ($SO(d) \simeq \mathbb{R}^d$) we know that $0 = \frac{d}{dt}\left(p_1 \cdot (q \times x_1) + p_2 \cdot (q \times x_2) + B(v_1)\omega_1 \cdot A_{v_1} q + B(v_2)\omega_2 \cdot A_{v_2} q \right)$ provided that $\forall Q \in SO(d)$ $W(q_1 - q_2, v_1, v_2) = W(Qq_1 - Qq_2, A(Q, v_1), A(Q, v_2))$.

EXAMPLE In the case of calamitic molecules we know that $A_{v_i} q$ is also given by the canonical action of $SO(d)$ in $\mathbb{R}^d$ and thus we have that

$$\frac{d}{dt}\left(q \cdot (x_1 \times p_1 + x_2 \times p_2 + v_1 \times B(v_1)\omega_1 + v_2 \times B(v_2)\omega_2) \right) = 0$$

Hence the conserved quantity is $x_1 \times p_1 + x_2 \times p_2 + v_1 \times B_1 \omega_1 + v_2 \times B_2 \omega_2$.

In the case of vanishing girth calamitic molecules we have, $B(\nu_i) = I$ we have that the conserved quantity is thus

$$\nu_1 \times \omega_1 + \nu_2 \times \omega_2 = \sum_{i=1}^{2}{}' (\nu_i \cdot \nu_i)\hat{\omega}_i - (\nu_i \cdot \hat{\omega}_i)\nu_i = \sum_{i=1}^{2} \mathrm{II}_i \hat{\omega}_i$$

where $\mathrm{II}_i = I - \nu_i \otimes \nu_i$ is precisely the inertia tensor of calamitic molecules with vanishing girth and $\hat{\omega}_i$ is the classical notion of angular velocity, i.e. $\hat{\omega}_i \times \nu_i = \omega_i$.

PROPOSITION Since the Lagrangian is a quadratic form of the conjugate momenta and is independent of t also the energy is conserved, i.e.

$$\frac{dH}{dt} = 0, \qquad H = \sum_{i=1}^{2}{}' \frac{1}{2} m |v_i|^2 + \frac{1}{2} \omega_i^T B(\nu_i) \omega_i + \mathcal{W}(x_1 - x_2, \nu_1, \nu_2).$$

We have thus used the MICROSTRUCTURE MANIFOLD to characterise the microscopic collisions. It is now time to derive a kinetic equation for complex fluids, we do so considering the N particle Hamiltonian, i.e.

$$\mathcal{H} = \sum_{i=1}^{N}{}' \frac{1}{2} m |v_i|^2 + \sum_{i=1}^{N}{}' \omega_i^T B(\nu_i) \omega_i + \sum_{1 \le i < j \le N} \mathcal{W}(q_i - q_j, \nu_i, \nu_j)$$

or alternatively in Hamiltonian coordinates,

$$\mathcal{H} = \sum_{i=1}^{N}{}' \frac{1}{2} m^{-1} |p_i|^2 + \sum_{i=1}^{N}{}' \frac{1}{2} \xi_i^T B(\nu_i)^{-1} \xi_i + \sum_{1 \le i < j \le N}{}' \mathcal{W}(q_i - q_j, \nu_i, \nu_j).$$

ξ_i: conjugate momenta to ν_i.

We know the Hamiltonian flow of this system must obey the Liouville equation, i.e.

$$\partial_t f_N = -\{f_N, \mathcal{H}\} \quad \text{with} \quad f_N = \sum_{i=1}^{N}{}' \delta(\Gamma_i - \Gamma_i(t)) \frac{1}{N}.$$

($\{\cdot,\cdot\}$: POISSON-BRACKETS.)

~~$\prod_{i=1}^{N} \delta(\Gamma_i - \Gamma_i(t))$~~ MISTAKE DURING TALK

Integrating the previous equation with respect to $\Gamma_{s+1}\Gamma_{s+2}\cdots\Gamma_N$ we obtain

$$\partial_t f_s + \{f_s, \mathcal{H}_s\} = \int \sum_{i=1}^{s}{}' \nabla_{p_i} f_{s+1} \cdot \nabla_{x_i} \mathcal{W}(x_i - x_{s+1}, \nu_i, \nu_{s+1}) + \sum_{i=1}^{s}{}' \nabla_{\xi_i} f \cdot \nabla_{\nu_i} \mathcal{W}(x_i - x_{s+1}, \nu_i, \nu_{s+1})$$

(f_s: marginal of f_N wrt $\Gamma_{s+1}\Gamma_{s+2}\cdots\Gamma_N$)

where $\mathcal{H}_s$ is the reduced Hamiltonian, $\mathcal{H}_s = \sum_{i=1}^{s} \frac{|p_i|^2}{2m} + \frac{1}{2}\xi_i^T B(v_i)^{-1}\xi_i + \sum_{1\le i<j\le s} W(\cdots)$ and we call this set of equations the BBGKY hierarchy.

Let us focus on the first two element of the BBGKY hierarchy, i.e.

$$\partial_t f_1 + \frac{p_1}{m}\cdot\nabla_{x_1} f_1 + B(v_1)^{-1}\xi_1\cdot\nabla_{v_1} f_1 = \int \nabla_{x_1} W(x_1-x_2, v_1, v_2)\cdot\nabla_{p_1} f_2 \, d\Gamma_2 + \int \nabla_{v_1} W(x_1-x_2, v_1, v_2)\cdot\nabla_{\xi_1} f_2 \, d\Gamma_2$$

$$\left.\begin{aligned} &\partial_t f_2 + \frac{p_1}{m}\cdot\nabla_{x_1} f_2 + B(v_1)^{-1}\xi_1\cdot\nabla_{v_1} f_2 + B(v_2)^{-1}\xi_2\cdot\nabla_{v_2} f_2 + \frac{p_2}{m}\cdot\nabla_{x_2} f_2 \\ &- \nabla_{x_1} W(x_1-x_2, v_1, v_2)\cdot(\nabla_{p_1} f_2 - \nabla_{p_2} f_2) \\ &- \nabla_{v_1} W(x_1-x_2, v_1, v_2)\cdot(\nabla_{\xi_1} f_2 - \nabla_{\xi_2} f_2) = 0 \end{aligned}\right| \quad (*)$$

$\nabla_{p_2} f_2$ and $\nabla_{\xi_2} f_2$: EXACT DIV when integrated.

At this point to obtain the Boltzmann equation is typical to introduce a change of coordinate to highlight the different length scales present in the previous equation, i.e.

$$\hat{x} = x_2 - x_1 \qquad \text{and} \qquad \hat{X} = \frac{1}{2}(x_1 + x_2).$$

We then observe that (*) can be rewritten in this coordinates as,

$$\partial_t f_2 + \frac{1}{2}\frac{p_1+p_2}{m}\cdot\nabla_{\hat{X}} f_2 + \underline{\underline{B}}(v_1)^{-1}\xi_1\cdot\nabla_{v_1} f_2 + \underline{\underline{B}}(v_2)^{-1}\xi_2\cdot\nabla_{v_2} f_2$$

$$\boxed{+ \frac{p_2-p_1}{m}\nabla_{\hat{x}} f_2 - \nabla_{x_1} W(x_1-x_2, v_1, v_2)\cdot(\nabla_{p_1} f_2 - \nabla_{p_2} f_2)}$$

$$- \nabla_{v_1} W(x_1-x_2, v_1, v_2)\cdot(\nabla_{\xi_1} f_2 - \nabla_{\xi_2} f_2) = 0$$

Observing that $\hat{x}$ varies on the microscopic length scales, while $\hat{X}$ varies on the mesoscopic scale we can isolate the boxed to be stationary with respect to the rest of the equation.

$$\frac{p_2-p_1}{m} = \nabla_{x_1} W(x_1-x_2, v_1, v_2)\cdot(\nabla_{p_1} f_2 - \nabla_{p_2} f_2)$$

Substituting this expression inside of BBGKY I we obtain a typical expression that is often a starting point to derive

the Boltzmann collision operator, i.e. $Q(f_2)$

$$\partial_t f_1 + \frac{p_1}{m}\cdot\nabla_x f_1 + B(v_1)^{-1}\xi_1\cdot\nabla_{v_1} f_1 = \int \frac{p_2-p_1}{m}\cdot\nabla_{\hat{x}} f_2 \, d\Gamma_2$$

$$+ \int \nabla_{v_1} W(x_1-x_2, v_1, v_2)\cdot\nabla_{\xi_1} f_2 \, d\Gamma_2$$

REMARK
Notice that to reach this point we have implicitely assumed that $\nabla_{v_i}\cdot$, $i=1,2$, varies on a different (larger) length then $\nabla_{\hat{x}}\cdot$.
WARNING The same trick can not be performed for $\nabla_v\cdot$, because we can't define $\hat{v}$ and $\hat{N}$. ⚠
It remains to understand how can we treat the term,

$$\int \nabla_{v_1} W(x_1-x_2, v_1, v_2)\cdot\nabla_{\xi_1} f_2 \, d\Gamma_2 \quad (**)$$

OPEN-PROBLEM How can we define fastly and slowly varying coordinates on $\mathcal{H}$?
One might think at first to assume that $f_2 = f_1 f_1$ which is the so called MEAN-FIELD MOLECULAR CAHOS ASSUMPTION.

Unfortunately even if this rewriting would make (**) more amenable would also lead to $Q(f_2)\equiv 0$, thus obtaining a REVERSABLE EQUATION. In fact the MEAN-FIELD MOLECUCAR CAHOS ASSUMPTION leads to the so called ~~LOSCHMIDT~~ LOSCHMIDT PARADOX and thus to a VLASOV type kinetic equation which is REVERSABLE in nature.

To this aim we observe that the MEAN-FIELD MOLECULAR CAHOS ASSUMPTION is motivated under interactions with long correlation range, so if we assume the order parameter varies on a length scales larger then the MEAN-FREE PATH or the COLISIONAL LENGTH SCALE, we can assume the interactions are weak only at the MICROSTRUCTURE level.
DEFINITION We say that a system, with 2-particle distributions $f(\Gamma_1,\Gamma_2,t)$ is subject to WEAK-ORDER INTERACTIONS if

$$\nabla_{\xi_i} f_2(\Gamma_i,\Gamma_j,t) = f_1(\Gamma_j,t)\,\nabla_{\xi_1} f_1(\Gamma_i,t).$$

Under this hypothesis we can rewrite the equation under consideration as,

$$\partial_t f_1 + \frac{p_1}{m}\cdot\nabla_{x_1} f_1 + B(v_1)^{-1}\xi_1\cdot\nabla_{v_1} f + \nu(x_1,v_1)\cdot\nabla_{\xi_1} f_1 = Q_B(f_1,f_1)$$

$$\nu(x_1,v_1) = -\int \nabla_{v_1} W(x_1-x_2, v_1, v_2)\, f_1(\Gamma_2, t)$$

The collision operator can be rewritten in Boltzmann form:

$$Q_B(f_1,f_1) = \int d\xi_2\, dp_2\, dv_2 \int d\theta\, d\varphi\; |p|\,\Sigma^{+}(|p|,\theta,\varphi)\, f_1(\Gamma_1',t)\, f_1(\Gamma_2',t)$$
$$- \int d\xi_2\, dp_2\, dv_2 \int d\theta\, d\varphi\; |p|\,\Sigma^{-}(|p|,\theta,\varphi) f_1(\Gamma_1',t)\, f_1(\Gamma_2',t)$$

Notice that this collision operator lacks the so called, which makes much harder to prove an H-THEOREM.

Following CERCIGNANI and LAMPIS RECIPROCITY BALANCE approach we can prove a BOLTZMANN-type INEQUALITY and an H-THEOREM.

THEOREM If we consider $\mathcal{H} = \int f_1 \log(f_1)\, d\xi_1\, dv_1\, dv_1$ and assuming that the system is subject to boundary conditions causing thermodynamic isolation, then $\frac{d\mathcal{H}}{dt} \leq 0$ as $t \to \infty$. Furthermore the "steady state" of the kinetic equation under consideration must be a Maxwellian of the form,

$$\bar{f}(\Gamma_1, t) = \exp\left[a(t) + b(t)\cdot p_1 + c(t)\,(A_{x_1} q + A_{v_1} q\cdot\xi_1) + d(t)\,(m^{-1}p_1\cdot p_1 + \xi_1^T B^{-1}\xi)\right]$$

for every $q \in SO(d)$ provided that A is TRANSITIVE.

OPEN PROBLEM What happens when A is not transitive.

CONJECTURE We conjecture that we still have a unique Maxwellian, fixed the moments, functions of the microscopic collision invariants.

DRAG TO EQUILIBRIUM

If we consider once again the case of calamitic molecules in two-dimensions and explicitely compute the gradients on the manifold we yield,

$$\partial_t f_1 + 2\hat{\omega}_1 \cdot \partial_{\theta_1} f_1 + \nu(\theta_1) \cdot \partial_{\hat{\omega}_1} f_1 = Q_{BGK} := -\frac{1}{\tau}(f - f^{(0)}) \quad (****)$$

under the assumption that f_1 is space homogeneous, i.e. $f = f(\theta_1, \hat{\omega}_1, \hat{v}_1, t)$, and we have replaced Q_B with a BGK type collision operator where $f^{(0)}$ is a Maxwellian of the form of the CURTISS – BOLTZMANN distribution, [CURTISS 1956]. We will also introduce the density $\pi(\theta_1, t) = \int dv_1 \int d\hat{\omega}_1 \, f(\theta_1, \hat{\omega}_1, v_1, t)$ and compute the continuity equation $\partial_t \pi + \partial_{\theta_1}(\pi\mu) = 0$ where

$$\pi\mu(\theta_1, t) := \int dv_1 \int d\hat{\omega}_1 \, f(v_1, \theta_1, \hat{\omega}_1, t).$$

Testing the equation $(****)$ with $\hat{\omega}_1$ and integrating we get,

$$\partial_t(\pi\mu) + \partial_{\theta_1} M - \nu(\theta_1)\pi(\theta_1, t) = \frac{\pi}{\tau}(\mu^{(0)} - \mu)$$

μ: STATIONARY VELOCITY

$\mu^{(0)}$: 0 (DRAG TO EQUILIBRIUM). This can be expressed by a Maxwellian with mean such that $\mu^{(0)} \equiv 0$

$$\mu = +\tau\nu(\theta_1) - \tau\pi^{-1}\partial_{\theta_1} M$$

Substituting this velocity in the continuity equation we have

$$\partial_t \pi + \partial_{\theta_1}(\pi\tau\nu(\theta_1)) - \tau\partial^2_{\theta_1} M = 0$$

we once again evaluate M using $M^{(0)}$ which we assume has fixed variance T, and thus $M^{(0)} = \int dv \int dw \, |w|^2 f$

$$= T\pi(\theta_1, t) \quad (8)$$

$$\partial_t \pi = \underbrace{zT}_{k} \partial^2_{\theta_1} \pi - z \partial_{\theta_1} (\pi v)$$

$$= k \partial^2_{\theta_1} \pi - z \partial_{\theta_1} \Big(\pi \underbrace{\partial_{\theta_1} \int d\omega_2 \int dv_2 \int d\hat{\omega}_2 - w(\theta_1, \theta_2) f_1(v_2, \hat{\omega}_2, \theta_2, t)}_{U(\theta_1, t)} \Big)$$

$$= k \partial^2_{\theta_1} \pi + z \partial_{\theta_1} (\pi \partial_{\theta_1} U(\theta_1, t))$$

$$= k \partial_{\theta_1} [\partial_{\theta_1} \pi + z \pi \partial_{\theta_1} (U(\theta_1, t))]$$

DOI - SMOLUCHOWSKI